

Tommaso de Martino

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WORK EXPERIENCE

European Central Bank ~ Statistics and Data Science Trainee

Feb 2026 – Present

- Build Python-based quantitative pipelines to compute supervisory Key Risk Indicators (KRIs) across large-scale banking datasets covering ECB-supervised institutions
- Integrate and validate Bloomberg market data for 100+ European banks, producing cross-sectional metrics used in market risk monitoring and financial analysis
- Collaborate directly with supervisors within the Single Supervisory Mechanism (SSM) to translate risk-monitoring objectives into quantitative analytical frameworks

EDUCATION

MS Economics and Finance (110/110 cum laude) ~ Sapienza University of Rome

Sep 2023 – Oct 2025

- Focus on quantitative methods, financial econometrics and machine learning in finance
- Merit scholarship: tuition fee waiver awarded to top 5% of cohort based on GPA
- Thesis: Asset Pricing with Machine Learning
Supervisor: Prof. Giuseppe Ragusa

BS Economics ~ Sapienza University of Rome

Sep 2020 – Oct 2023

- Focus on Economic Theory
- Thesis: Dedollarization of the International Monetary System
Supervisor: Prof. Marco Di Pietro

PROJECTS

Asset Pricing with Machine Learning

- Applied machine learning methods to a large-scale equity dataset of 3M+ firm-month observations covering 23,729 firms and 807 predictors to forecast cross-sectional stock returns
- Neural networks achieved the strongest out-of-sample predictive performance across the model set
- Built long-short equity portfolios based on model predictions, with the best equally weighted strategy delivering 28.2% annualized return

Volatility Forecasting ~ Euro Stoxx 50

- Benchmarked GARCH against machine learning models including MLP, LSTM and Random Forest to forecast Euro Stoxx 50 volatility on 3,763 daily observations
- Engineered 72 cross-asset features across FX, rates, equity indices and commodities to improve volatility forecasting performance
- The best-performing MLP achieved 9% lower out-of-sample RMSE than the GARCH benchmark

Macro Forecasting with Factor Models

- Extracted 7 latent macroeconomic factors from 126 U.S. economic time series (1959–2025) to model broad macroeconomic dynamics
- Produced rolling real-time forecasts for key macro indicators including industrial production and CPI inflation
- The factor model achieved out-of-sample RMSE of 1.37 for industrial production and 0.60 for CPI

SKILLS & INTERESTS

Technical

- Python (advanced): Pandas, Polars, NumPy, scikit-learn, TensorFlow, PyTorch
- R (advanced); Tableau (proficient)
- Experience working with large-scale datasets and server-based computing environments
- Git / GitHub for version control and project documentation

Languages

Italian (native); English (fluent); Spanish (intermediate); French (intermediate)

Interests

Financial markets, Muay Thai and nature